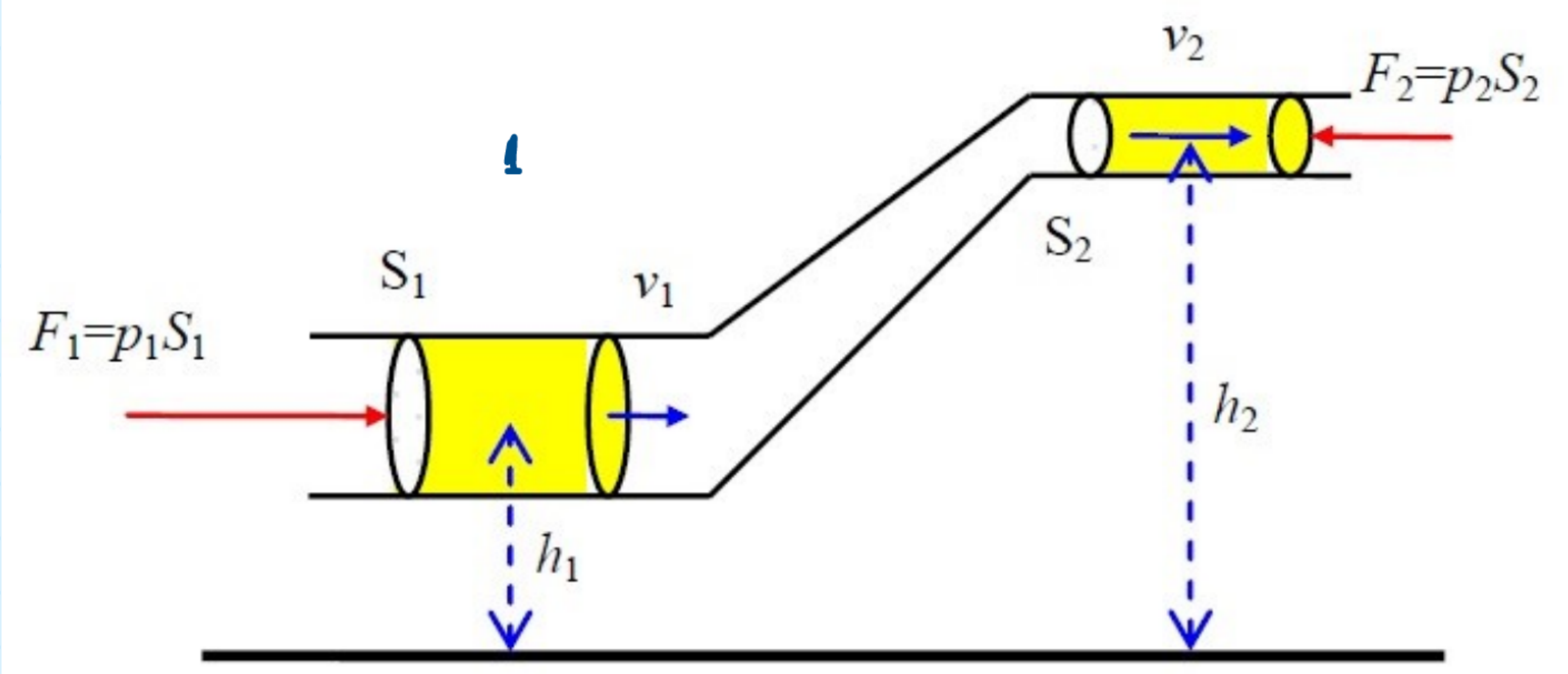
Ligjerata 8d

Ekuacioni i Bernulit

$A = E_2 - E_1 \quad *$

$E_1 = E_{k_1} + E_{p_1} = \frac{1}{2}mv_1^2 + mgh_1$

$E_2 = E_{k_2} + E_{p_2} = \frac{1}{2}mv_2^2 + mgh_2$



$A = F_1 l_1 + F_2 \cdot l_2 = p_1 S_1 \cdot v_1 t - p_2 S_2 v_2 t \quad *$

$A = \frac{1}{2}mv_2^2 + mgh_2 - \frac{mv_1^2}{2} - mgh_1 \quad **$

$\frac{1}{2}mv_2^2 + mgh_2 - \frac{1}{2}mv_1^2 - mgh_1 = p_1 S_1 v_1 t - p_2 S_2 v_2 t$

$-\frac{1}{2}mv_1^2 - mgh_1 - p_1 S_1 v_1 t = -\frac{1}{2}mv_2^2 - mgh_2 - p_2 S_2 v_2 t \quad / (-1)$

$\frac{mv_1^2}{2} + mgh_1 + p_1 S_1 v_1 t = \frac{mv_2^2}{2} + mgh_2 + p_2 v_2 S_2 t$

$m = \rho V$

$\left.\begin{array}{l} V_1 = S_1 v_1 t \\ V_2 = S_2 v_2 t \end{array}\right\}$ $\quad Q = \frac{V}{t} = \frac{S l}{t} = S \cdot v \quad \frac{V}{t} = S v \,/ \cdot t$

$\boxed{V_1 = V_2} \quad \boxed{V = S \cdot v \cdot t}$

$\frac{mv_1^2}{2} + mgh_1 + p_1 V = \frac{mv_2^2}{2} + mgh_2 + p_2 \cdot V \quad / : V$

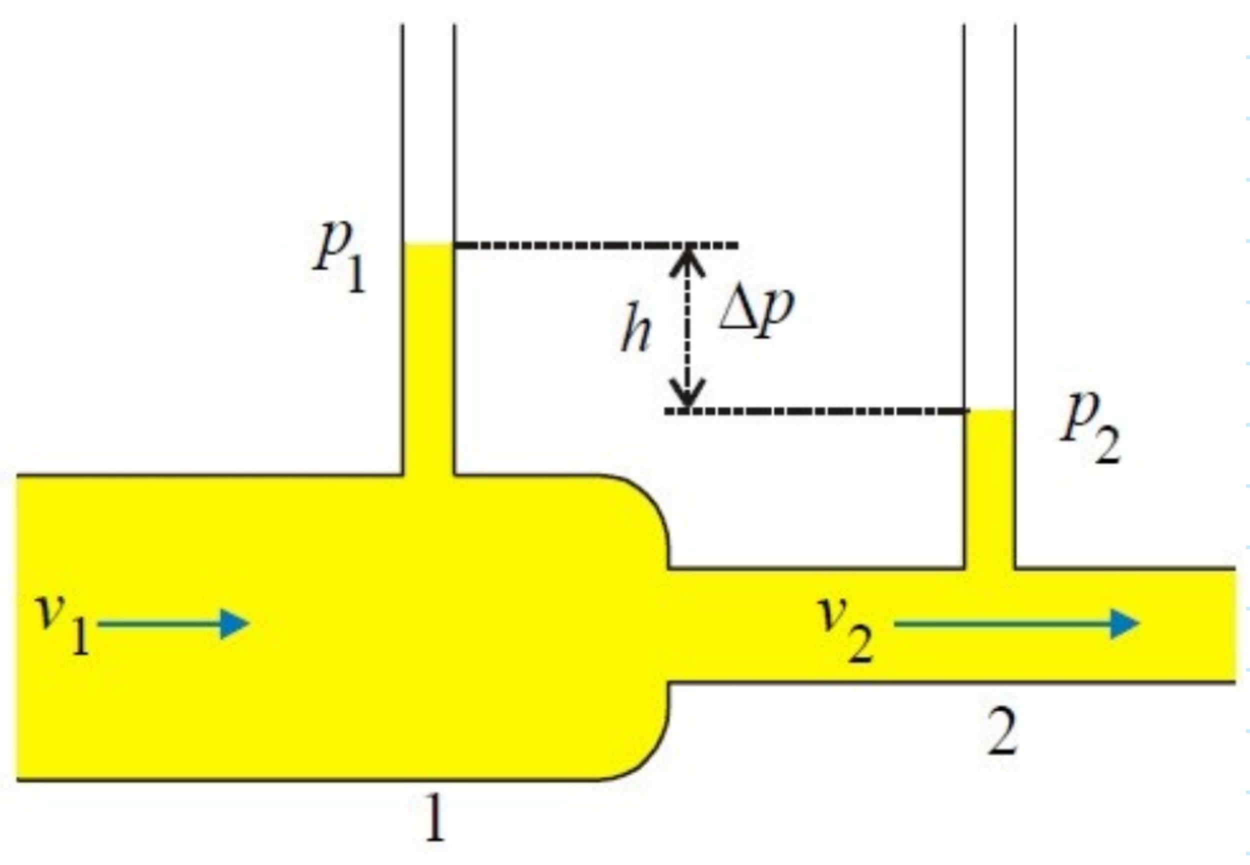
$\frac{m}{V}\frac{v_1^2}{2} + \frac{m}{V} \cdot gh_1 + p_1 = \frac{m}{V} \cdot \frac{v_2^2}{2} + \frac{m}{V} gh_2 + p_2$

$\frac{\rho v_1^2}{2} + \rho g h_1 + p_1 = \frac{\rho v_2^2}{2} + \rho g h_2 + p_2$

$\boxed{\frac{\rho v^2}{2} + \rho g h + p = \text{konst.}}$

$\frac{\rho v_1^2}{2} + \underbrace{\rho g h_1}_{} + p_1 = \frac{\rho v_2^2}{2} + \underbrace{\rho g h_2}_{} + p_2$

$h_1 = h_2$



$p_1 + \frac{\rho v_1^2}{2} = p_2 + \frac{\rho v_2^2}{2}$

$p_1 - p_2 = \frac{\rho v_2^2}{2} - \frac{\rho v_1^2}{2}$

$\Delta p = \frac{\rho}{2}(v_2^2 - v_1^2)$

$\underline{\Delta p = \rho g h}$

$\cancel{\rho} g h = \frac{\cancel{\rho}}{2}(v_2^2 - v_1^2)$

$\boxed{2gh + v_1^2 = v_2^2}$